

Key Points:

- Oceanic Kelvin wave (OKW) response per westerly wind event (WWE) forcing in models is stronger than the observed
- Models have too-thin and overly stratified mixed layers compared to observations, which may lead to overly strong OKW responses
- The mixed layer biases are consistent with the net surface heat flux between the atmosphere and the ocean

Supporting Information:

Supporting Information may be found in the online version of this article.

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Mixed Layer and Oceanic Kelvin Wave Response Biases to Equatorial Pacific Westerly Wind Events in CMIP6 Models

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Abstract This study explores the biases in westerly wind events (WWEs) and their induced oceanic Kelvin waves (OKWs) in seven CMIP6 models from three modeling centers that provide daily thermocline depth output. Among the WWEs and OKWs identified in historical simulations, WWEs are generally weaker than observed for a given OKW response, suggesting that OKWs in models respond too strongly to WWE forcing. The equatorial ocean mixed layers in the models are generally too thin and overly stratified compared to observations. These biases may help inhibit the turbulence-induced mixed layer deepening and trap the WWE-provided momentum within the ocean surface layers. Thus, they support an exaggerated sea surface height buildup and thermocline deepening in OKW initiation regions, which leads to an overly strong OKW response. These biases may be associated with a too-strong net heat flux between the atmosphere and the ocean.

Plain Language Summary Oceanic Kelvin waves (OKWs) play a significant role in regulating the upper-ocean heat distribution and impact coupled ocean-atmosphere phenomena in the tropics such as El Niño-Southern Oscillation (ENSO). Addressing biases in OKWs and their atmospheric forcing by westerly wind events (WWEs) could help improve current ENSO predictions and ENSO-induced weather and climate forecasts. This study reveals that climate models tend to have a weaker westerly wind forcing than what is observed in the real world, although the strength of the upper ocean OKW response is the same. This suggests that the models may overestimate how strongly OKWs respond to WWEs. One possible reason for this bias is the models' too-thin and too-stable ocean surface mixed layers, which can hinder the effective exchange of energy and mass between the surface and deeper layers. This could cause the energy from WWEs to remain trapped in the surface layers rather than being transferred deeper. As a result, the sea surface height and thermocline depth respond more than they should in the regions where OKWs are initiated.

1. Introduction

Equatorial oceanic Kelvin waves (OKWs) play a significant role in regulating upper ocean thermal distributions and mediating atmospheric and ocean phenomena across scales throughout the tropics (e.g., Bergman et al., 2001; Delcroix et al., 1991; Harrison & Schopf, 1984; Kessler & McPhaden, 1995; Kutsuwada & McPhaden, 2002; McPhaden & Yu, 1999; Rydbeck et al., 2019; Rydbeck & Jensen, 2017; Webber et al., 2010; Wyrski, 1977; Zhang & Gottschalck, 2002). For instance, OKWs serve as a key link between higher-frequency atmospheric variability such as the intraseasonal timescale Madden-Julian oscillation (MJO; Madden & Julian, 1971, 1972) over the Indo-Pacific warm pool (e.g., Chen et al., 2016, 2017; Jauregui & Chen, 2024; Puy et al., 2016) and the inter-annual timescale El Niño-Southern Oscillation (ENSO) in the tropical Pacific. Specifically, synoptic-scale low-level westerly wind anomalies that accompany organized convective cloud systems are a large source of westerly wind events (WWEs) over the western Pacific (e.g., Fasullo & Webster, 2000; Feng & Lian, 2018; Puy et al., 2016; Seiki & Takayabu, 2007). The WWEs impart strong and long-lasting wind stresses on the ocean surface, which can generate downwelling OKWs on the equator (e.g., Battisti, 1988; Delcroix et al., 1991; Kessler et al., 1995). The eastward-propagating OKWs influence El Niño onset and evolution by deepening the thermocline, raising sea surface height (SSH) and increasing upper ocean heat content as they traverse the central-to-eastern equatorial Pacific (Kessler et al., 1995; McPhaden & Yu, 1999; Puy et al., 2019; Zhang & Gottschalck, 2002). In turn, ENSO cycles regulate lower-frequency background conditions that alter large-scale sea surface temperature (SST) patterns and atmospheric circulations (Bjerknes, 1969), which further modulate surface wind anomalies (e.g., Hendon et al., 1999; Jin, 1997; Kessler, 2001; Pohl & Matthews, 2007; Tam & Lau, 2005) and subsequent OKWs.

As a communicator between wind variability and ENSO, OKWs provide a link between higher-frequency atmospheric anomalies (i.e., WWEs) and lower-frequency ENSO evolution. OKWs, therefore, play a major role in ENSO forecasts and subseasonal-to-seasonal forecasts of ENSO-induced weather and climate anomalies (Seo & Xue, 2005). Thus, faithfully representing WWE amplitude and frequency, air-sea coupling in OKW initiation, and OKW amplitude and propagation are key to accurate ENSO representations in both simplified (e.g., Chen et al., 2022; Chen & Fang, 2023; Wang et al., 2025) and state-of-the-art dynamical models (e.g., Ji et al., 2023, 2025). Since models in Coupled Model Intercomparison Project (CMIP) are a principal tool for understanding ENSO variability and climate change; it is natural to ask to what extent CMIP models can represent WWEs, OKWs, and WWE-OKW coupling.

Feng and Lian (2018), Lian et al. (2018), and Tan et al. (2020) showed varying degrees of overestimation or underestimation of WWE occurrence in CMIP5 and other individual models. More recently, Riley Dellaripa et al. (2024, RD24 hereafter) evaluated the frequency, location, amplitude, duration, and zonal extent of WWEs over the equatorial Pacific in 30 CMIP6 models. They found that model WWEs are less frequent and weaker than observed, which is partially related to underpredicted MJO and convectively coupled Rossby wave variance in models. The WWE features in models partially determine their OKW responses. However, to date, there is limited work on the evaluation of OKW characteristics across climate models. This is primarily because of the historical lack of availability of daily ocean fields in CMIP archives that are needed to resolve intraseasonal OKWs. Recently, daily-resolved 20°C isothermal depth was released by some CMIP6 models, which allows the analysis of OKW variance. Cui et al. (2024, C24 hereafter) diagnosed OKW features in the equatorial Pacific in CMIP6 models, which revealed remarkably slow OKW propagation in many CMIP6 models, consistent with their seasonally late El Niño onset shown in Planton et al. (2021).

The OKW biases may originate from biased WWE forcing or biased air-sea coupling processes. Previous studies have shown that models have sometimes struggled with coupling between WWEs and OKWs and their induced SST evolution (e.g., Hayashi & Watanabe, 2019; Lengaigne et al., 2003). In one study, Lengaigne et al. (2004) prescribed a strong WWE in a ten-member coupled model ensemble to examine if an El Niño event could be triggered. Six members failed due to misrepresented air-sea coupling processes during the early development of the OKWs, partly stemming from the weak air-sea feedback set by biases in the seasonality of the trade wind and the ocean's vertical thermal structure. In another study, Peng et al. (2019) reproduced the 2017 extreme coastal El Niño initiated by successive WWEs in the MIT general circulation model. However, the sea level anomalies associated with the WWE-OKW coupling were initially weaker and more confined to the western Pacific. Thus, a more general assessment of WWE-OKW coupling in models is needed.

In this study, we expand upon the work of RD24 and C24 and evaluate the fidelity of the coupling between WWEs and the OKW responses at the OKW initiation stage in CMIP6 models. This study aims to understand the OKW responses to WWE forcing in the equatorial Pacific and scale interactions in CMIP6 models and their potential biases by analyzing horizontal features of OKWs, their coupling characteristics with WWEs, and their relationship to ocean mean state conditions.

2. Data and Methods

2.1. Reanalysis Data

TropFlux daily zonal and meridional wind stresses, τ_x and τ_y , respectively (Praveen Kumar et al., 2013), were employed for WWE identification and feature analysis (RD24). The Global Ocean Reanalysis and Simulations (GLORYS)-12v1 product (Jean-Michel et al., 2021) was used to analyze observed OKWs. With this product, we derived the 20°C isothermal depth (T20d), which serves as a proxy for the thermocline depth, from daily temperature fields for comparison with the same variable provided in the CMIP6 models.

When exploring the upper ocean dynamics at the OKWs initiation stage, we used the daily mixed layer depth (MLD) from the Estimating the Circulation and Climate of the Ocean Version 4 Release 4 (ECCO V4r4) product (ECCO Consortium et al., 2021). Four other monthly MLD products from LOPS laboratory and ECMWF Ocean Reanalysis System 5 (ORAS5; Zuo et al., 2019) were also used, and their details are introduced in Section 2.3. Monthly meridional velocity (v) from ORAS5 was examined to ensure consistency with CMIP6 models, as daily v is not available in the CMIP6 archive. The MLD and v are used as references to infer potential biases in wind-induced mixed layer (ML) variations in the models. Additionally, the GLORYS monthly averaged ocean potential

temperature and salinity were used to derive vertical profiles of the mean Brunt-Väisälä frequency, N^2 , to diagnose upper ocean mean stratification during OKW initiation.

Other monthly data used to explore the sources of ML biases include outgoing longwave radiation (OLR) from National Oceanic and Atmospheric Administration (NOAA) interpolated OLR product (Liebmann & Smith, 1996), precipitation from the Global Precipitation Climatology Project (GPCP) v2.3 (Adler et al., 2003), and 10-m wind speed from the fifth generation ECMWF reanalysis (ERA5; Hersbach et al., 2023).

The above reanalyses were all regridded to a $1^\circ \times 1^\circ$ latitude-longitude grid. The variables mentioned above are referred to as observations for convenience, with the caveat that they are observationally constrained reanalyses rather than true observations. The period of all atmosphere and ocean fields used in this study is 1993–2014, since the satellite-based ocean reanalysis is only available after 1993, and the CMIP6 historical simulations end in 2014.

2.2. CMIP6 Models

Following C24, the seven CMIP6 member models that provide daily T20d in the historical simulations were examined. They are IPSL-CM6A-LR, IPSL-CM6A-LR-INCA, MPI-ESM-1-2-HAM, MPI-ESM1-2-HR, MPI-ESM1-2-LR, NorESM2-LM, and NorESM2-MM. These models employ atmosphere, ocean, and other components as detailed in C24.

Other ocean fields analyzed from the CMIP6 archive include daily wind stress and daily MLD and monthly vertical profiles of potential temperature, salinity, and ocean meridional velocity. The CMIP model variables were regridded to the same $1^\circ \times 1^\circ$ grid as observations, and the period of 1993–2014 was analyzed. In this study, only ensemble member r1i1p1f1 of each CMIP6 model was used. A single-member analysis is commonly applied for intermodel evaluations when the internal variability across ensemble members within a specific model is not the focus.

2.3. MLD Definitions

MLD is estimated by the depth over which the physical parameters of the ocean state (e.g., temperature, salinity, and density) are well mixed and homogeneous. A common approach for MLD estimation is to define a threshold for temperature or density difference from its value at a reference depth. The depth where the threshold is met is the MLD. The ECCO V4r4 MLD product used in this study follows Kara et al. (2000), defining MLD by a 0.8°C temperature threshold, whereas the MLD in CMIP6 protocols is derived by a density criterion of a 0.125 kg m^{-3} increase (Griffies et al., 2016; Heuzé, 2021; Heuzé et al., 2023), with both referenced to a depth of 10 m. Both are larger thresholds than the commonly used 0.5°C temperature threshold or 0.01 and 0.03 kg m^{-3} density thresholds employed by de Boyer Montégut et al. (2004). A larger threshold likely results in a deeper MLD estimate (de Boyer Montégut et al., 2004; Suga et al., 2004). As a result, differing MLD definitions in CMIP6 models and the ECCO product complicate direct comparison. To address this uncertainty, we incorporated four additional data sets with varying MLD definitions to evaluate CMIP6 MLD biases relative to the ECCO V4r4 product. These include monthly MLD from ORAS5 with the definition of a 0.01 and 0.03 kg m^{-3} density increase from 10-m-depth (abbreviated as ORAS5_D001 and ORAS5_D003) and climatological MLDs from LOPS laboratory (de Boyer Montégut et al., 2004) defined by a 0.2°-C decrease from the surface (dBM_T02) and by an equivalent density increase corresponding to a 0.2°-C temperature decrease (dBM_EqT02), respectively. The latter two products were derived from in situ measurements of temperature and salinity profiles and are thus closer to true observations. Note that the MLD criteria employed in these products are generally stricter than those in CMIP6 protocols.

2.4. WWE and OKW Response Identifications

Following RD24, our analysis uses equatorially averaged (2.5°S – 2.5°N) 120-day high-pass filtered zonal wind stresses τ_x over the tropical Pacific (120°E – 100°W) to identify strong WWEs capable of exciting OKWs. The filtering emphasizes higher-frequency variations, since longer timescale τ_x is already balanced by the upper ocean zonal pressure gradient and is unable to initiate OKW anomalies (McPhaden & Taft, 1988; McPhaden & Yu, 1999).

In RD24, when τ_x anomalies have a magnitude of at least 0.04 N m^{-2} , lasting for at least 5 days, and have a zonal extent of at least 10° longitude, it is considered a WWE. A value of 0.04 N m^{-2} is approximately two standard

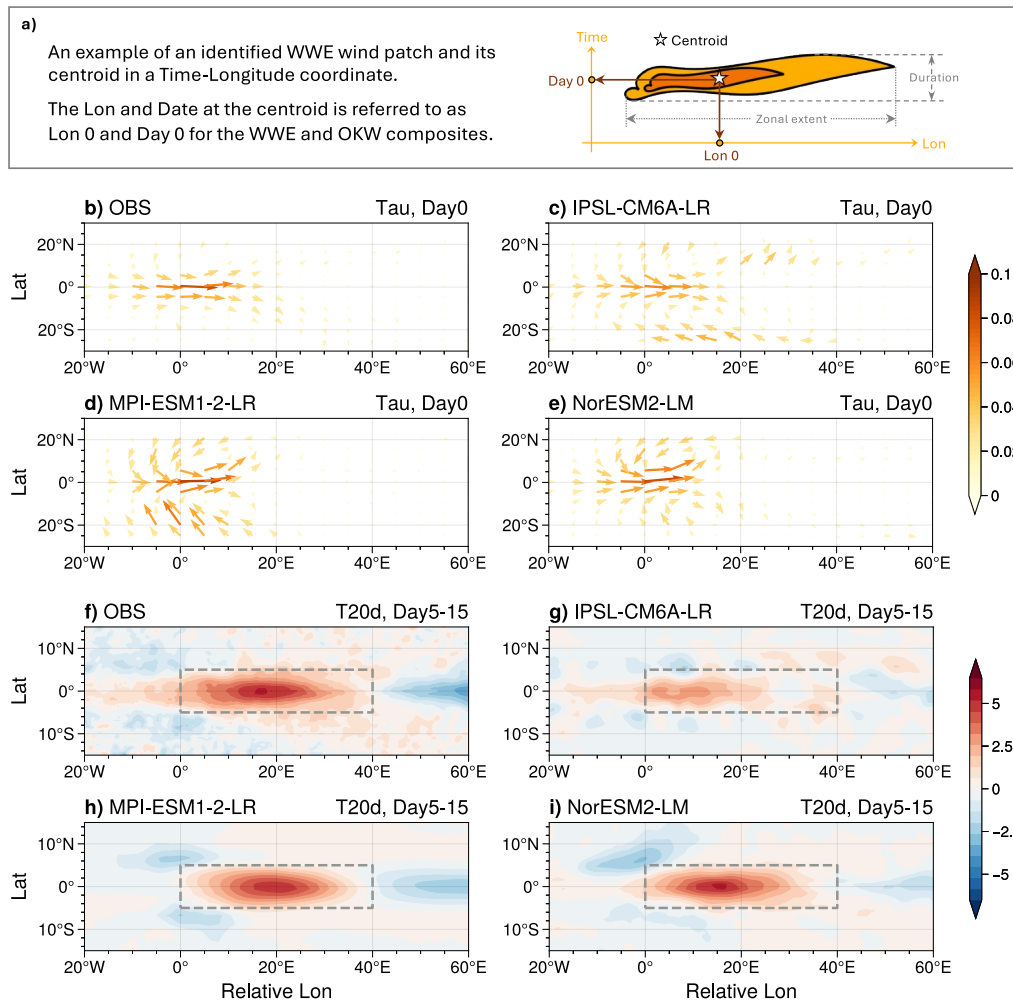


Figure 1. (a) An example of the identified westerly wind event (WWE) wind patch, and its Lon 0 and Day 0. (b)–(e) the identified WWE wind stress (N m^{-2}) on Day 0 in the observations and three models from each model family as examples, and (f)–(i) the corresponding oceanic Kelvin wave (OKW) responses represented by T20d anomalies (meters) averaged over Day 5–15 in the observations and the three models. The dashed boxes indicate the OKW initiation region used for OKW amplitude calculation. Note that the patterns in (b)–(i) are composited based on the region spanning from 20° west to 60° east to the Lon 0 in each WWE, so the x-axis is the relative longitude referenced to the Lon 0 rather than the true longitude, whereas the y-axis is the true latitude. The colors in (b)–(e) indicate the wind stress magnitudes.

deviations (2σ) above the equatorially averaged mean τ_x in the TropFlux observations. To increase our sample size for statistical purposes, we reduced the τ_x magnitude threshold to 1σ . This allowed for weaker but still anomalously strong WWEs to be considered. The 1σ threshold of about 0.02 N m^{-2} is similar among models and observations. The number of identified WWE occurrences in most models is close to the observed, which is 315 on average; but the occurrences are fewer in MPI-ESM1-2-HR (215). Causes of the reduced WWE occurrence in this model are not clear but do not appear to be linked to the biased MJO variability (RD24). Other biases in synoptic variability over the tropical Pacific (e.g., tropical cyclones and cold surges from midlatitudes) may contribute to the WWE bias (e.g., Yu et al., 2003; Yu & Rienecker, 1998).

For each identified WWE patch in longitude-time space (Figure 1a), the longitude and the date at the τ_x -weighted centroid define the relative longitude and the day of the corresponding OKW initiation (Lon 0 and Day 0, respectively). Accordingly, for one WWE, days after Day 0 are represented by positive values, and longitudes east or west of Lon 0 are referred to as Lon E or Lon W, respectively. For example, the day lagged 2 days to Day 0 is Day 2 and the location 2° west of Lon 0 is Lon 2°W .

The OKW response to the WWE is assessed using the daily T20d anomaly, which was obtained by removing the mean annual cycle and then filtering with a 30–120-day band-pass filter. We defined the period from Day 5 to Day 15 as the OKW initiation stage because OKWs are typically excited by accumulated wind stresses for successive days (e.g., Giese & Harrison, 1990). The region of Lon 0 to Lon 40°E and 5°S to 5°N is defined as the OKW initiation region (boxes in Figures 1f–1i). Thus, the OKW responses hereafter refer to the T20d anomalies averaged over the initiation stage and initiation region.

3. Spatial Structure of the Identified WWEs and OKWs

Figures 1b–1i show the composite spatial structures of the WWEs on Day 0 and OKW responses in the observations and models. Only three model results are shown because the WWE and OKW patterns are similar between models within each model family. In observations, the wind stresses are most prominent on the equator, with a maximum magnitude of about 0.1 N m^{-2} near Lon 0 that diminishes to the east and west. The models show a similar τ_x zonal extent and zonal distribution on the equator. However, model WWEs consistently display a pronounced Rossby gyre-like pattern, with cyclonic circulations on both sides of the equator, which is not observed. This indicates that model WWEs may be more strongly associated with Rossby waves, which is also suggested by the too-strong Rossby signals in the coherence spectra between wind stress and T20d anomalies in models (Figure S1 in Supporting Information S1).

Figures 1f–1i show the horizontal patterns of the OKW responses at the initiation stage (Day 5–15 average). Observations and the NorESM and MPI models exhibit T20d anomalies with a zonal scale of about 40° longitude and a maximum T20d anomaly of about 5–6 m. In contrast, the IPSL models produce weaker, smaller-scale OKW responses, primarily because their WWEs are displaced toward the central-eastern Pacific (RD24), where conditions are less favorable for deep convection and coupled westerlies. Additionally, overly strong tropical instability waves in IPSL models (not shown) may further dampen the ocean response, resulting in overall weaker OKW amplitudes.

The influence of WWE wind stress on the ocean depends not only on the wind stress magnitude but also on the duration and zonal extent of the event. To capture these combined effects, we adopted the concept of integrated wind work (IWW), defined as the sum of τ_x over the entire time-longitude patch of each WWE (shaded area in Figure 1a). This metric is conceptually similar to the “wind measure index” in Harrison and Vecchi (1997) and the “wind event index” in Puy et al. (2016). By construction, IWW incorporates the wind stress magnitude, duration, and zonal extent—factors to which the oceanic response is generally proportional (Giese & Harrison, 1990). Thus, IWW represents the total wind forcing exerted by a WWE on the ocean and provides a basis for examining the statistical connections between individual WWEs and their OKW responses, as analyzed in the following section.

4. Stronger Model OKW Responses per WWE Forcing

4.1. Overly Strong OKW Responses in Models

Figures 2a–2h show the probability distribution of the OKW T20d responses at the initiation stage (Day 5–15 average). Compared to the observations, the models have a higher probability of weak OKWs (i.e., T20d anomaly <2 m), as seen in the differences of the heights of bars for each T20d bin. Additionally, most models have a longer tail than observed, indicating a higher frequency of high-amplitude OKWs than observed. IPSL models are an exception, consistent with their weak mean OKW response seen in Figure 1g.

The color shading of the bars in Figures 2a–2h indicates the mean IWW associated with T20d anomaly bins. Darkening colors from left to right indicate increasing IWW with increasing T20d anomalies. This pattern suggests that the OKW amplitudes increase with increasing IWW in all the models and observations. However, a key result from this figure, as indicated by the downward-pointing arrows over T20d bins for models, is that the model IWWs for a given T20d bin are nearly all smaller than observations, especially for the weaker OKWs (i.e., T20d bins on the left). This implies that a weaker WWE in models could excite an OKW of comparable amplitude to observations. There are exceptions. For example, in the observations, the average IWW is only 10.4 N m^{-2} for the 4–5-m T20d bin, which is an outlier to the overall trend of IWWs increasing from left to right. This leads to the result that the model IWWs for the 4–5-m T20d bins overestimated (i.e., upward-pointing arrows). In general,

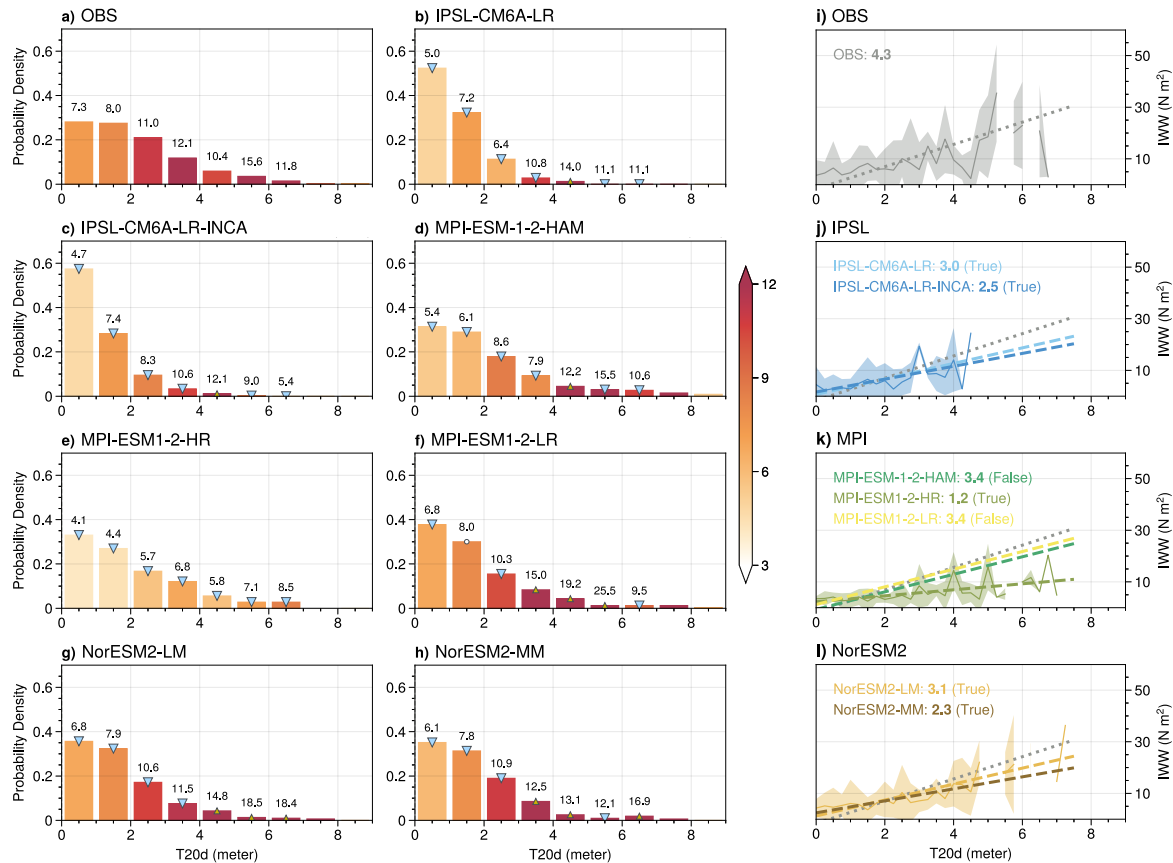


Figure 2. (a–h) Probability density distributions of the oceanic Kelvin wave T20d amplitude in the observed and model westerly wind events (WWEs), with the color shading and the number over each bin showing the average integrated wind work (IWW) amplitude of the WWEs within the bin. The upward (or downward)-pointing triangles over each bin show larger (or smaller)-than-observed IWW for the corresponding T20d bin. (i–l) The statistical distribution of the IWWs binned by the T20d amplitudes in the observations and three example models. T20d bin intervals are 0.25 m. Thin curves represent the median IWW values for each bin. Shading represents one standard deviation above and below the median. Bold dashed lines are the least squares polynomial fit for the curves, with the slopes shown in the upper left corner. The True in the parentheses means the corresponding model fit coefficient is significantly different from the observations at the 0.1 significance level by F test. Different colors represent different models and are text-marked on each panel. The gray dashed line in (i)–(l) is the same, indicating the observation result.

however, OKWs of a given amplitude in models are associated with weaker wind forcing than in observations, indicating an overly strong OKW response per WWE forcing.

To further verify the model's overly strong OKW responses relative to observations, we applied the least squares polynomial fit to the median IWW within each bin versus the T20d bin central value to obtain their statistical relationships. The results are plotted in the right panels of Figure 2, with the fit plotted as dotted (observations) or dashed (models) lines and the spread of IWW values (i.e., $\pm 1\sigma$) shown with shading. The observations and all models show an increase of IWW with T20d, consistent with the gradually darker color of the probability distributions. However, the models display a lower rate of increase than observations, as seen with the dashed colored (models) and gray dotted (observations) lines in Figures 2i–2l, respectively, and by their slopes listed in the upper left corners. As determined using an F test, most of the models show a significantly different increasing rate than observed at a significance level of 0.1 (as indicated with “True” in Figure 2). This indicates that, for a given amplitude of T20d responses, the model IWW is smaller than observed, consistent with the idea that the model OKWs respond too strongly to wind forcing.

4.2. Ocean MLD and Stratification Biases

Ocean dynamics communicate the effects of ocean surface forcing by WWEs to the thermocline. Specifically, WWEs generate equatorial eastward surface currents (the Yoshida jet; Yoshida, 1959), upper ocean Ekman

meridional convergence, and downwelling on the equator (Cronin & McPhaden, 2002; Lengaigne et al., 2004), which together raise SSHs and depress T20d (Zebiak & Cane, 1987) along the equator. The upper ocean Ekman convergence transports waters meridionally toward the equator, and stronger convergence and Yoshida jet activity are associated with intensified downwelling, contributing to SSH elevation and thermocline suppression.

The magnitude of these oceanic responses depends not only on WWE intensity but also on the ocean stratification, particularly the ML, which modulates the surface current and convergence efficiency. For a given MLD, a stronger WWE (i.e., larger IWW) will produce a stronger surface zonal current and upper ocean meridional Ekman convergence and a larger SSH and T20d displacement. Conversely, for a given WWE, a thinner ML will yield a larger SSH and T20d anomaly, as surface momentum fluxes trapped within a thin ML will generate a stronger surface current than momentum fluxed into a thicker ML (Vialard & Delecluse, 1998a, 1998b; Vialard et al., 2002). This stronger surface current can further sustain and strengthen the upper ocean meridional convergence. Hence, MLD biases can lead to OKW amplitude biases (e.g., Hackert et al., 2019).

In addition to ML thickness, ML stability can also influence the OKW response. MLs typically deepen following WWEs via shear-driven turbulent mixing at the ML base (e.g., Zhang & McPhaden, 2000; Zhang & Rothstein, 1999). Shear-driven deepening of the ML is favored when the density gradient across the ML base is weak and is inhibited when the gradient is strong, as the energy required to mix the water masses is proportional to the density stratification (e.g., Huang, 1998). Hence, a more stratified ML base would inhibit the deepening of the ML and help sustain the zonal surface currents and upper ocean convergence that generate the OKW SSH and T20d displacement.

Given the likely dependence of OKW amplitude on MLD and stability, we assessed the mean MLD and Brunt-Väisälä frequency (N^2 ; a measure of stability) at the ML base (Figures 3a and 3b). Figure 3a shows the mean MLD averaged over the equatorial Pacific for models and for several observational or reanalysis products. MLs in CMIP6 models, which are defined using a 0.125 kg m^{-3} density criterion referenced to a depth of 10 m, are thinner than the ML defined by a 0.8°C temperature threshold from the ECCO product. Model MLs are also thinner than MLs defined by other temperature (0.2°C ; dBM_T02) or temperature equivalent density thresholds (dBM_EqT02) and are comparable to those estimated by either a 0.01 or a 0.03 kg m^{-3} density threshold (ORAS5_D001 or D003, also referenced to a depth of 10 m). The latter four thresholds, however, are much smaller (stricter) than the 0.125 kg m^{-3} threshold used in models. Because density increases with depth, applying the same 0.125 kg m^{-3} threshold to those reanalyses or observations would yield MLs much thicker than those in models (de Boyer Montégut et al., 2004; Suga et al., 2004). These comparisons collectively indicate that the CMIP6 models have a too-thin mean ML, which could amplify the OKW responses to WWEs.

To further understand the nature of MLD biases and their relation to WWEs, Figures 3c–3f show the MLD distributions during the OKW preinitiation stage (Day -15 to -5), representing the preexisting state for ocean responses. All models exhibit a narrower distribution centered near 25 m, whereas observations display a broader one around 55 m. This systematically shallower ML is consistent with Figure 3a and likely contributes to the amplified OKW responses in the models.

Figure 3b displays the mean ocean stratification, N^2 , averaged over a 30-m layer centered on the ML base for the months when WWEs occur. The models overestimate ML base stratification by 10%–50% compared to observations. Figures 3g–3j further illustrate that model N^2 distributions prior to WWE events are skewed toward larger values, consistent with the biases in mean values in Figure 3b. This suggests that the MLs in models, which are already too thin compared to observations, may also be too resistant to deepening by vertical turbulent mixing during WWEs. The too-strong stratification bias in models may help maintain the too-thin ML bias during WWEs and thus sustain conditions that are overly supportive of a strong surface response to WWEs.

The deviations in N^2 between models and observations (Figures 3g–3j) are less pronounced than those in MLD (Figures 3c–3f), suggesting that the OKW response biases may be primarily driven by MLD biases, with secondary contributions from excessive stratification at the ML base. Moreover, it is also worth noting that the too-thin and overly stratified ML might be related to a higher-order baroclinic OKW response with stronger surface height and zonal current responses (Giese & Harrison, 1990). This is consistent with the biased higher-order baroclinic OKWs seen in C24.

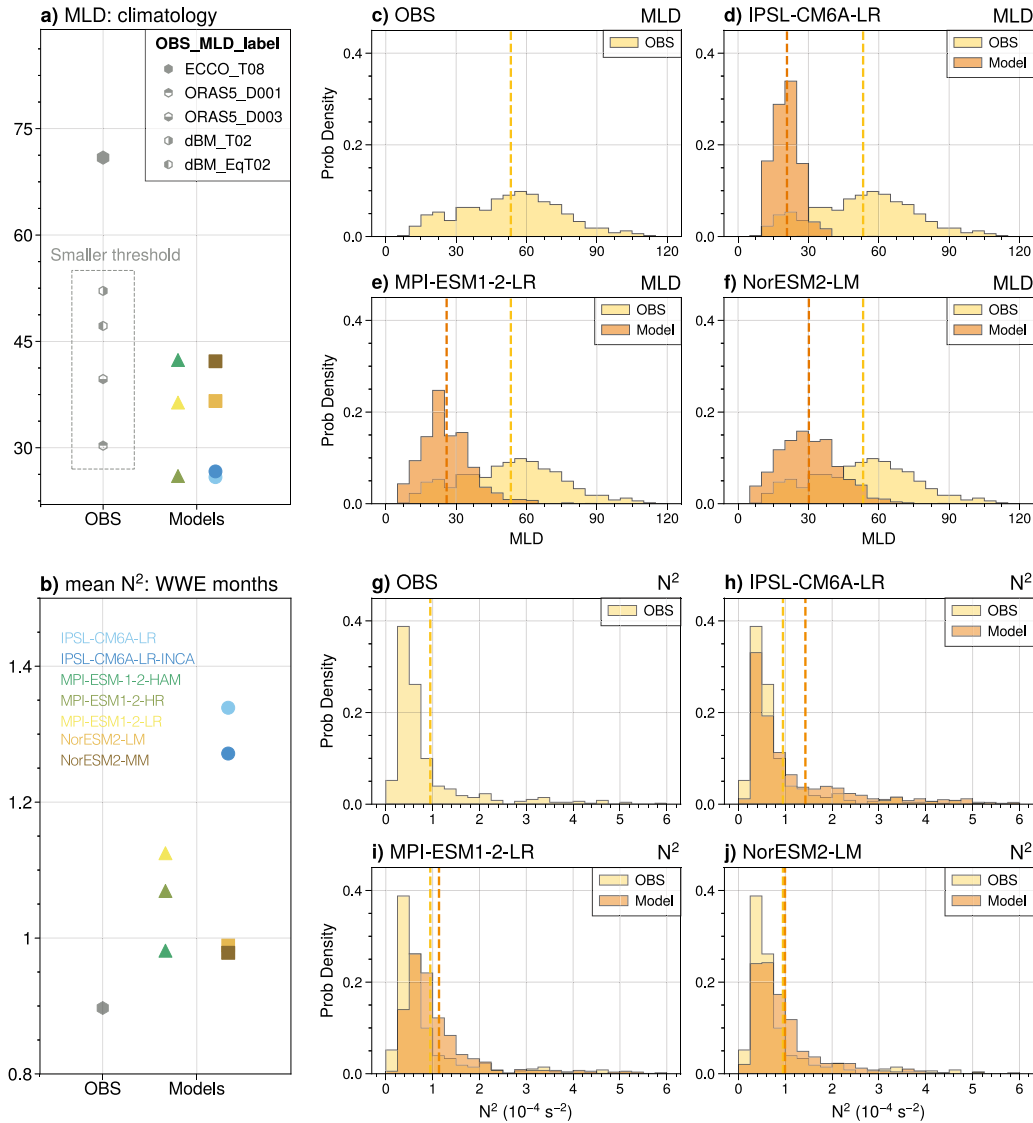


Figure 3. (a) Climatological mixed layer depth (MLD; meters) averaged over 2°S – 2°N , 150°E – 120°W , and (b) mean Brunt-Väisälä frequency N^2 (10^{-4} s^{-2}) averaged within $\pm 15 \text{ m}$ of the mixed layer (ML) base in the oceanic Kelvin wave initiation regions for the months containing westerly wind event (WWE) Day 0. The half-filled markers are the observational MLD references listed in Section 2.3, which were derived from smaller (stricter) MLD thresholds. Panels (c)–(f) show probability density distributions of daily MLD averaged from Day -15 to -5 in the OKW region, representing preexisting MLD conditions prior to WWEs. Panels (g)–(j) show corresponding distributions of N^2 at the ML base for the months containing WWE Day 0, in the observations and three example models.

To explore these ideas further, we examined the observed and simulated changes in MLD anomaly (ΔMLD hereafter) from OKW preinitiation (Day -15 to -5) to OKW initiation (Day 5 – 15) to indicate the MLD deepening associated with the WWE occurrences and OKW initiations. Daily MLD anomalies were computed using the same method for obtaining daily T20d anomalies (Section 2.4). Figures 4a–4d display the distributions of ΔMLD binned by T20d anomalies at the OKW initiation region. In observations, the averaged ΔMLD is 3.3 m (marked by the gray dot in Figure 4a), whereas the models show smaller or negative values. This is consistent with the overly stratified ML base in models (Figures 3b and 3g–3j) because a more stable ML base would inhibit ML deepening through turbulent mixing. Furthermore, in Figure 4a, the observations show a clear positive correlation such that a stronger OKW response generally corresponds to a stronger MLD deepening, as both are responses to

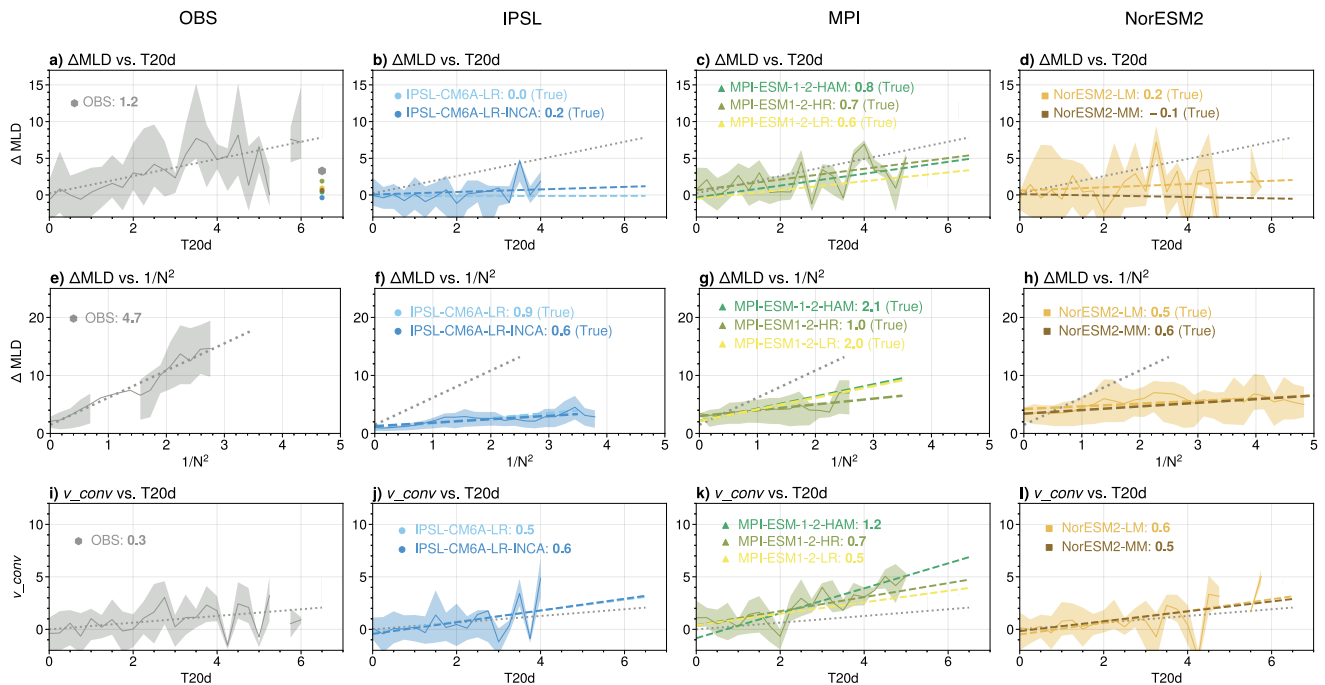


Figure 4. Statistical distributions of the OKW region-averaged (a–d) mixed layer depth (MLD) anomaly changes (ΔMLD), which were derived from the daily MLD field and represent the changes between Day 5–15 and Day –15 to –5 and (i–l) monthly meridional velocity convergence averaged over the top 50 m and Lon 20°W–60°E, binned by the T20d amplitudes in the observations and three example models. T20 bin intervals are 0.25 m. The middle row panels (e–h) are similar to (a–d) but for ΔMLD binned by the $1/N^2$ at the mixed layer base (± 15 m) for westerly wind events (WWEs). The thin curves, bold dashed lines, shadings, numbers, and Trues at the upper left are the same as in Figure 2. The markers at the lower right in (a) are averaged ΔMLD across all WWEs in individual models and observations.

WWE forcing. However, such a connection is underestimated in all models, with all models displaying a significantly smaller slope than observed at a 0.01 significance level.

These underestimated connections between ΔMLD and T20d anomalies suggest that the overly strong OKW responses in models may be a result of an overly stratified upper ocean in models that inhibits turbulence-induced ML deepening. This interpretation is supported by the relationship between ΔMLD and N^2 at the ML base. Figures 4e–4h show that ΔMLD generally increases with decreasing N^2 (i.e., increasing x -axis value) in both observations and models, consistent with the fact that a weakly stratified ML base would favor turbulence-induced ML deepening. The ΔMLD increase rate is significantly larger in the observations, indicating that models require a much smaller N^2 (i.e., a less stable ML base and a larger x -axis value) to achieve the same ΔMLD as observed. This further supports that the overly stratified ML base in models suppresses ML deepening and contributes to the biased OKW responses.

Evidence from the upper ocean meridional convergence also supports the above reasoning. Figures 4i–4l show the anomalous monthly meridional velocity convergence (v_{conv}) averaged over the top 50-m layer (roughly the ML thickness) binned by T20d anomalies. Due to the unavailability of the daily velocity field in CMIP6 outputs, the convergence is the difference of the monthly v anomaly between 5°S and 0° and 0° and 5°N over Lon 20°W–60°E, where the v anomaly is the monthly v value with the seasonal cycle removed. All the models show a stronger-than-observed v_{conv} anomaly for a given T20d anomaly. The larger v_{conv} agrees with the fact that model MLs are too thin and too stable at their bases compared to observations (Figures 3a and 3c–3f). The overly stratified upper ocean layers reduce the vertical eddy viscosity, leading to a shallower Ekman layer (Ekman, 1905; Price & Sundermeyer, 1999) and a more surface-confined, intensified upper ocean meridional Ekman convergence. Meanwhile, the meridional convergence can be further sustained and strengthened by the strong wind-induced zonal currents. These collectively further promote sea surface growth. It is worth noting that the stronger v_{conv} in models during WWE months is not simply a reflection of climatological biases, as the models exhibit different signs and magnitudes of mean state divergence biases compared with observations (Figure S2 in Supporting Information S1). The stronger-than-observed mean divergence in some models (i.e., the two IPSL models,

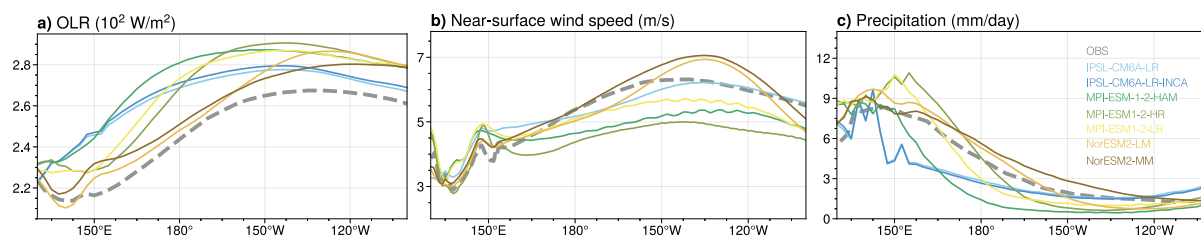


Figure 5. The climatological (a) outgoing longwave radiation, (b) near-surface (10-m) wind speed, and (c) precipitation averaged over 5°S–5°N from 1993 to 2014.

MPI-ESM1-2-HAM, and MPI-ESM1-2-LR) may partly contribute to their shallower mixed layers, consistent with the observed link between enhanced surface-layer divergence or upwelling and a reduced mixed layer depth (e.g., Wang & McPhaden, 1999).

It is worth noting that the overly strong oceanic response to WWEs in models persists across different seasons and ENSO phases. Although both WWEs and MLDs exhibit pronounced seasonal and state-dependent variations, which are key factors in ENSO phase locking and predictability (Lopez & Kirtman, 2014), the biases in modeled ocean responses to WWEs and MLD remain robust, systematic, and largely insensitive to these variations.

5. Discussions

Our analysis has revealed overly strong OKW responses per WWE forcing in CMIP6 models. The models have thinner MLs and an overly stratified ML base compared to the observations. These findings are consistent with Ekman convergence being confined to a shallower layer in models that contribute to the overly responsive sea surface and OKWs.

A natural next question is what leads to the too-thin ML and stratification biases in the models? Multiple processes may contribute to overly strong buoyancy in the ML: too-strong net heating of the ocean surface, either through enhanced solar heating or reduced surface sensible and latent heat fluxes or excessive freshening by rainfall. In addition, biases in the ocean turbulent mixing parameterization could also lead to ML biases.

To explore possible causes for MLD biases in models, simulated and observed equatorial OLR, precipitation, and near-surface wind speed are shown in Figure 5. In Figure 5a, the models show a positive mean OLR bias across the equatorial Pacific, indicating less high cloudiness in the models. This implies that excessive shortwave radiation reaches the ocean surface, particularly when high cloud biases are not offset by compensating low cloud biases (Ceppi et al., 2024; Miao et al., 2021). The resulting surface heating may thin and stabilize the MLs. Additionally, most models have weaker-than-observed wind speed over the western Pacific where WWEs often occur (Figure 5b), which would contribute to weaker-than-observed ocean surface cooling by surface sensible and latent fluxes (e.g., Li et al., 2020) and weaker generation of upper ocean turbulence that can deepen the ML. Since these surface fluxes act to cool and destabilize the upper ocean, their reduction would limit their ability to destabilize and deepen the ML. In Figure 5c, precipitation biases are mixed. The positive precipitation biases over the warm pool in some models could further stabilize the model MLs through salinity stratification. These findings suggest that atmospheric biases are at least partly responsible for the MLD biases, which contribute to OKW amplitude biases. Additionally, internal ocean dynamics may also play an important role in the MLD biases, as deficiencies in vertical mixing, submesoscale processes, and large-scale circulation can precondition the upper ocean structure.

6. Summary

This study aims to provide a process-based understanding of model biases in WWE-OKW relationships, which is essential for accurately representing OKWs and ENSO evolution in ocean-atmosphere coupled models. We found biases in WWE-OKW relationships at the OKW initiation stage in seven CMIP6 models that provide daily thermocline depth (T20d) output, which is used to quantify OKW characteristics. We found that models have varying degrees of overestimation or underestimation of the mean WWE and OKW amplitude (Figure 1) and weaker-than-observed WWE forcing per unit amplitude of OKW T20d response (Figure 2). Together, these findings imply overly strong OKW responses to WWEs in the models.

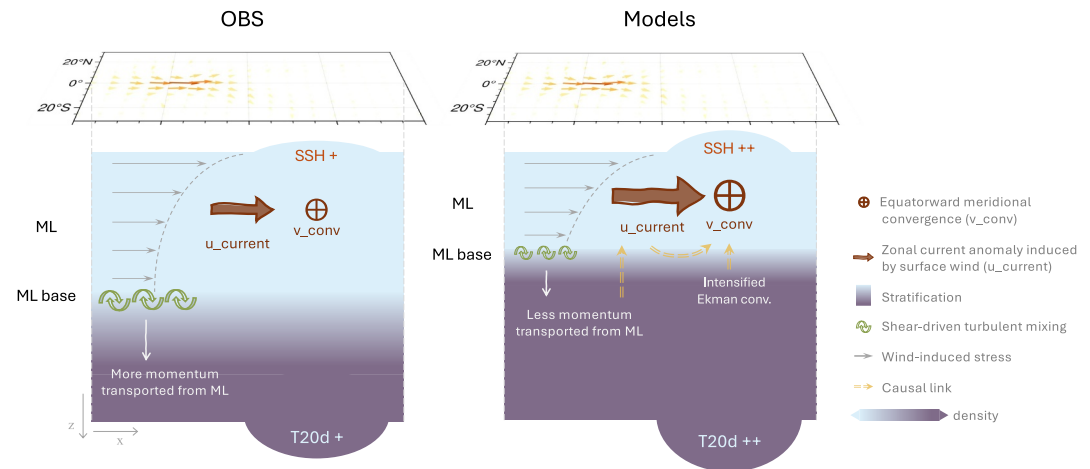


Figure 6. Conceptual schematic illustrating how mixed layer biases in CMIP6 models influence the oceanic response to westerly wind events. The vertical cross-section depicts ocean processes across 5°S to 5°N.

The too-strong OKW response to WWEs appears to be linked to a too-thin and overly stratified ML in models (Figure 3). These biases would help sustain a stronger equatorial zonal surface current and stronger near-surface meridional Ekman convergences onto the equator than the observed, thus favoring a stronger SSH increase and thermocline deepening in the OKW initiation region (Figure 4). Figure 6 conceptually summarizes the above processes. The thin ML bias and the overly stratified ML base are likely associated with the net heat flux biases between the atmosphere and the ocean, such as the shortwave radiative heating and latent heat flux and, for some models, excessive precipitation over the warm pool (Figure 5). Future studies are needed to detangle how the net heat flux biases affect the upper ocean variations.

Several limitations of this study warrant discussion. Due to the lack of daily ocean velocity, temperature, and salinity output in CMIP6 models, our analysis relies on monthly velocity fields and derived monthly buoyancy frequency to approximate WWE-induced upper ocean variations and the regulatory role of buoyancy. This constrains our ability to examine upper ocean dynamics in detail, particularly the causal relationships among daily fluctuations in SSH, MLD, and thermocline structure. Additionally, discrepancies between the MLD definitions used in CMIP6 models and those in reanalysis products may further hinder the robustness and accuracy of our results. A more quantitative diagnosis of how mixed layer characteristics modulate equatorial wave responses and associated upper ocean variability will require model simulations that provide daily ocean output.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

The climatological MLD dBM_T02 and dBM_EqT02 in Figure 3a were downloaded from the LOPS laboratory at TropFlux zonal wind stress data (Praveen Kumar et al., 2013) and are available at https://mld.ifremer.fr/Surface_Mixed_Layer_Depth.php. The daily ocean temperature and salinity in Global Ocean Reanalysis and Simulations (GLORYS)-12v1 product (Jean-Michel et al., 2021) are available at <https://data.marine.copernicus.eu/products>. The monthly velocity and MLD (ORAS5_D001 and D003) from ECMWF Ocean Reanalysis System 5 (ORAS5) reanalysis (Zuo et al., 2019) are provided by Copernicus Climate Change Service (C3S) at <https://cds.climate.copernicus.eu/datasets/reanalysis-oras5?tab=download>. Estimating the Circulation and Climate of the Ocean Version 4 Release 4 (ECCO V4r4) MLD was downloaded at https://podaac.jpl.nasa.gov/dataset/ECCO_L4_MIXED_LAYER_DEPTH_05DEG_DAILY_V4R4#. The CMIP6-member model outputs (Bentsen et al., 2019; Boucher et al., 2018, 2021; Jungclaus et al., 2019; Neubauer et al., 2019; Seland et al., 2019; Wieners et al., 2019) in this study were downloaded from The Earth System Grid Federation at https://esgf-metagrid.cloud.dkrz.de/sea_rch/cmip6-dkrz/https://cerweb.ifremer.fr/deboyer/mld/Data.php. National Oceanic and Atmospheric Administration interpolated OLR product provided by the NOAA PSL, Boulder, Colorado, USA is available at <https://psl>.

noaa.gov/data/gridded/data.olrcdr.interp.html. Precipitation from GPCP v2.3 was downloaded at <https://psl.noaa.gov/data/gridded/data.gpcp.html>. Near-surface wind speed was from ERA5 (Hersbach et al., 2023) at <https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels-monthly-means?tab=download>.

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